

OLS versus SPTLS, by hand: a worked counterpoint on a rotating series

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Companion note to *Symbolic Pure Time* (F-000)

Abstract

This note fits one and the same 20-point series twice — once by ordinary least squares (OLS) on a lag, as a review, and once by Symbolic Pure-Time Least Squares (SPTLS), as exposition. The series is a rotation in phase space: a single-lag scalar model cannot represent it, while the SPTLS one-step map can. At every step we show two things side by side: *(i)* how to reproduce the arithmetic by hand — which entry of which matrix multiplies what — and *(ii)* which object each method is really using. The two columns are a running counterpoint: OLS rests on the autocovariance (Toeplitz) matrix, the SVD solve, and the Fourier/power-spectrum reading; SPTLS rests on the kinematic embedding, the polar decomposition, and the Plancherel-graded characteristic function (PGCF). Every number is reproduced by `replication.py`.

1 The data

The series is generated by a damped rotation in a two-dimensional state, observed through one coordinate: $s_{t+1} = \rho R(\omega) s_t + \text{noise}$ with $\rho = 0.97$, $\omega = 0.62$, and $z_t = (s_t)_1$. It oscillates with a slow decay (rounded to three decimals, so the reader can recompute):

$$z_1, \dots, z_{20}: \quad \begin{array}{cccccccccccccccccccc} 1.000, & 0.790, & 0.315, & -0.265, & -0.693, & -0.839, & -0.631, & -0.218, & 0.233, & 0.590, \\ & 0.699, & 0.526, & 0.211, & -0.199, & -0.482, & -0.583, & -0.426, & -0.150, & 0.185, & 0.437. \end{array}$$

Figure 1 shows why a scalar lag will struggle: in the phase plane $(z_t, \Delta z_t)$ the trajectory *rotates*, and a single real coefficient can only scale, never rotate.

2 Method 1 — OLS on a lag (review)

We fit $z_t = c + \varphi z_{t-1} + \varepsilon_t$ by ordinary least squares.

What you compute by hand. Stack the $n = 19$ rows $X = [\mathbf{1}, z_{t-1}]$ and the target $y = z_t$. The normal matrix and its solution are

$$X^\top X = \begin{pmatrix} 19 & 0.063 \\ 0.063 & 5.486 \end{pmatrix}, \quad \hat{\theta} = (X^\top X)^{-1} X^\top y = \begin{pmatrix} c \\ \varphi \end{pmatrix} = \begin{pmatrix} -0.029 \\ 0.741 \end{pmatrix}.$$

Each entry of $X^\top X$ is a sum over the sample: the $(2, 2)$ entry is $\sum_t z_{t-1}^2 = 5.486$, the $(1, 2)$ entry is $\sum_t z_{t-1}$, the $(1, 1)$ entry is the row count. The one-step fit gives $R^2 = 0.647$ (RMSE 0.295).

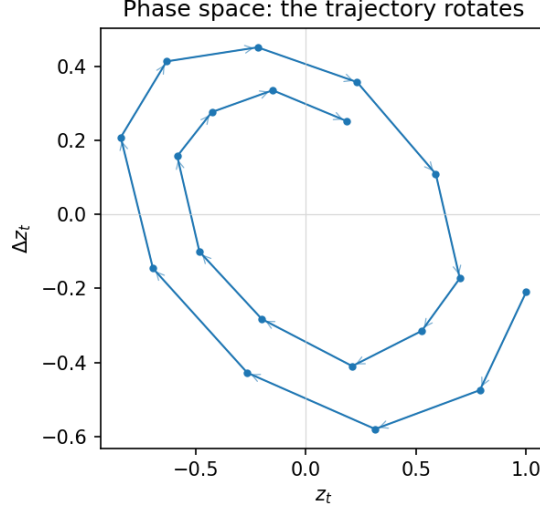


Figure 1: The series in phase space $(z_t, \Delta z_t)$: the trajectory spirals inward. A one-lag scalar model multiplies by a single real number and cannot produce rotation; an operator on the embedding can.

What is really being used.

- **The autocovariance matrix.** $X^\top X$ is (up to centring and scale) the sample autocovariance matrix of the series. It is *Toeplitz* — constant along diagonals — because it assumes time-translation invariance, and its coordinates are raw past values (a Cartesian lag basis).
- **The SVD as solver.** In practice one does not invert $X^\top X$; one takes the singular value decomposition $X = U\Sigma V^\top$ and forms $\hat{\theta} = V\Sigma^+U^\top y$. The SVD is the numerically stable *solver* of the least-squares problem.
- **The Fourier reading.** The autocovariance matrix is diagonalised by the Fourier transform; its spectral content is the power spectrum (Wiener-Khinchin). This is the natural spectral analysis attached to OLS-on-lags.

The single coefficient $\varphi = 0.741$ can only damp; it cannot turn the trajectory, so the fit lags the oscillation badly (Figure 2, left).

3 Method 2 — SPTLS on the embedding (exposition)

SPTLS fits the one-step map of the kinematic embedding $\mathbf{q}(t) = (z_t, \Delta z_t, \Delta^2 z_t) \in \mathbb{R}^3 \subset \text{Cl}(3, 0)$ by the matrix $\hat{M}$ with $\mathbf{q}(t+1) \approx \hat{M} \mathbf{q}(t)$.

What you compute by hand. Form the embedding for $t \geq 3$, then the two sums over the transitions, $C_0 = \sum_t \mathbf{q}(t)\mathbf{q}(t)^\top$ and $C_1 = \sum_t \mathbf{q}(t+1)\mathbf{q}(t)^\top$, and solve $\hat{M} = C_1 C_0^{-1}$:

$$C_0 = \begin{pmatrix} 3.862 & 0.667 & -1.329 \\ 0.667 & 1.924 & 0.367 \\ -1.329 & 0.367 & 0.665 \end{pmatrix}, \quad \hat{M} = \begin{pmatrix} 0.350 & 1.156 & -0.679 \\ -0.650 & 1.156 & -0.679 \\ -0.650 & 0.156 & -0.679 \end{pmatrix}.$$

$\hat{M}$ is markedly non-symmetric, $\|\hat{M} - \hat{M}^\top\|/\|\hat{M}\| = 1.25$ — the fingerprint of a rotation. Reading it through its polar decomposition $\hat{M} = RP$:

$$R = \begin{pmatrix} 0.318 & 0.941 & 0.115 \\ -0.168 & 0.175 & -0.970 \\ -0.933 & 0.289 & 0.213 \end{pmatrix} \text{ (a rotation by } 98.4^\circ\text{),} \quad P = \begin{pmatrix} 0.926 & -0.095 & 0.318 \\ -0.095 & 1.487 & -0.690 \\ 0.318 & -0.690 & 0.899 \end{pmatrix}.$$

The one-step fit gives $R^2 = 0.999$ (RMSE 0.016).

What is really being used.

- **The kinematic embedding.** The design coordinates are the instantaneous *jet* $(z, \Delta z, \Delta^2 z)$ — position, velocity, acceleration — not lagged values. The Gram C_0 is the covariance of the jet, and the fitted $\hat{M}$ is the discrete flow.
- **The polar decomposition.** $\hat{M}$ is solved by the same SVD as OLS; the polar decomposition $\hat{M} = RP$ is read *from* that SVD ($R = UV^\top$, $P = V\Sigma V^\top$) *after* the fit. It separates the abelian stretch P (the scalar gains, what OLS would see) from the non-abelian rotor R (grade-2) — the rotation a scalar coefficient cannot hold.
- **The PGCF.** The probabilistic layer is the Plancherel-graded characteristic function, $\mathbb{E}[\exp(\xi \wedge X)]$ valued in $\text{Spin}(3) \cong SU(2)$. The power spectrum is its abelian ($j = 0$) shadow; the higher graded components are the non-abelian extension. (Distributional; developed in the companion F-001.)

4 The counterpoint, term by term

The two methods solve the same kind of least-squares problem, but through different objects. The contrast is the point of the note:

	OLS on lags	SPTLS
Coordinates	lagged values $z_{t-1}, z_{t-2}, \dots$ (Cartesian, backward)	kinematic jet $(z, \Delta z, \Delta^2 z)$ (forward state)
Gram matrix	autocovariance, <i>Toeplitz</i> (time-translation inv.)	embedding covariance C_0 (no Toeplitz structure)
Unknown	coefficient <i>vector</i> $\hat{\theta}$	<i>operator</i> $\hat{M} \in \text{Cl}(3, 0)$
Solver	SVD / QR of the design	SVD / QR of the design (<i>same</i>)
Post-hoc reading	— (a vector has none)	polar $\hat{M} = RP$: stretch P + rotor R
Spectral layer	Fourier $\rightarrow$ power spectrum ($U(1)$)	PGCF $\rightarrow$ Plancherel tower ($SU(2)$); power spectrum is the $j=0$ shadow
Can rotate?	no (scalar gain)	yes (rotor R)

Two entries deserve emphasis. First, the *solver* is the same: SVD does the least squares in both. What differs is that SPTLS solves for an operator, so its SVD admits a polar reading (rotation $\times$ stretch) that a coefficient vector does not. Second, the spectral objects are nested, not rival: the OLS power spectrum is exactly the abelian shadow of the SPTLS PGCF.

5 Fits and performance

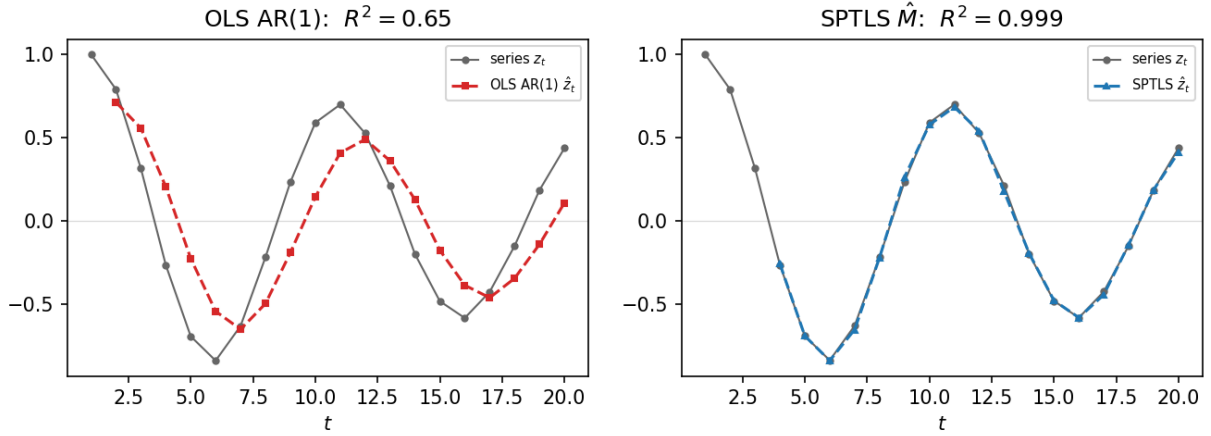


Figure 2: One-step predictions. Left: OLS AR(1) — the single coefficient damps but cannot turn, so the prediction lags the cycle ($R^2 = 0.65$). Right: SPTLS $\hat{M}$ — the rotor tracks the rotation ($R^2 = 0.999$).

Estimator	RMSE (one-step)	R^2	note
OLS AR(1)	0.295	0.647	one lag, scalar gain — cannot rotate
OLS AR(2)	0.023	0.998	two lags, linear — recovers the oscillation
SPTLS $\hat{M}$	0.016	0.999	embedding + rotor, one step

An honest note. The oscillation here is *linear*, so ordinary least squares on *two* lags (AR(2)) also recovers it: the gap above is between SPTLS and OLS *at one lag*. The point of the comparison is therefore representational, not a universal superiority claim. What SPTLS adds, already at one step, is that the rotation is made explicit — it is the rotor R of $\hat{M}$ — where the scalar-lag model must reach for higher order and never names the rotation as such. On genuinely nonlinear, regime-switching dynamics (the Lorenz butterfly of F-000, Appendix C) the gap is not closed by adding lags.